

n = and $U = 01$ - = not

Exercise No. 2.2 :-

1 to 4, 16, 17, 18, 19

Question No. 1 :-

Let A be the set of students who live within one mile of school and let B be the set of students who walk to classes. Describe the students in each of these sets.

(a)

$A \cap B$

The set of students who live within one mile of school and the set of students who walk to classes.

(b)

$A \cup B$

The set of students who live within one mile of school or the set of students who walk to classes.

(c)

$A - B$

The set of students who live within one mile of school but do not the set of students who walk to classes.

(d)

$$B - A$$

The set of students who walk to classes but do not the set of students who live within one mile of school.

Question 10 2:-

Suppose that A is the set of Sophomores at your school and B is the set of students in discrete mathematics at your school. Express each of these sets in terms of A and B .

set (a)

the set of Sophomores taking discrete mathematics in your school.

$$A \cap B$$

$$\cap \cup \cup$$

(b)

the set of Sophomores at your school who are not taking discrete mathematics.

$$A - B$$

(c)

the set of students at your school who either are Sophomores or are taking discrete mathematics.

$$A \cup B$$

(d)

the set of students at your school who either are not sophomores or are not taking discrete mathematics.

$$A^c \cup B^c$$

Union = Arrange Difference = A element not in B.
Intersection = Same.

Question 10 3 -

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find

(a)

$$A \cup B$$

$$A = \{1, 2, 3, 4, 5\} \cup B = \{0, 3, 6\}$$

$$A \cup B = \{0, 1, 2, 3, 4, 5, 6\}$$

(b)

$$A \cap B$$

$$A = \{1, 2, 3, 4, 5\} \cap B = \{0, 3, 6\}$$

$$A \cap B = \{3\}$$

(c)

$$A - B$$

$$A = \{1, 2, 3, 4, 5\} - B = \{0, 3, 6\}$$

$$= \{1, 2, 4, 5\}$$

(d)

$$B - A = \{0, 3, 6\} - A = \{1, 2, 3, 4, 5\}$$

$$B - A = \{0, 6\}$$

Question No 4:-

$$A = \{a, b, c, d, e\}$$

$$B = \{a, b, c, d, e, f, g, h\}$$

(a)

$A \cup B$

$$A = \{a, b, c, d, e\} \cup B = \{a, b, c, d, e, f, g, h\}$$

$$A \cup B = \{a, b, c, d, e, f, g, h\}$$

(b)

$A \cap B$

$$A = \{a, b, c, d, e\} \cap B = \{a, b, c, d, e, f, g, h\}$$

$$A \cap B = \{a, b, c, d, e\}$$

(c)

$A - B$

$$A = \{a, b, c, d, e\} - B = \{a, b, c, d, e, f, g, h\}$$

$$A - B = \{\phi \text{ empty set}\}$$

(d)

$B - A$

$$B = \{a, b, c, d, e, f, g, h\} - A = \{a, b, c, d, e\}$$

$$B - A = \{f, g, h\}$$